

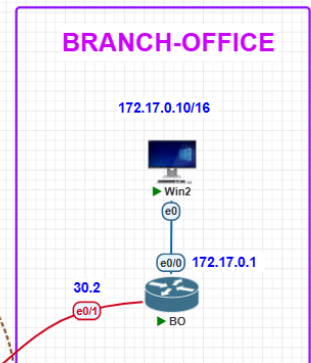
LAB 3 PORT FORWARDING

GURUCHANDHRAN M

TASK 1: NAT CONFIGURATION IN BRANCH OFFICE ROUTER

PALOALTO PORT-FORWARDING

```
BO
NVI0 172.17.0.1 YES unset up up
BO#int e0/0
^
% Invalid input detected at '^' marker.
BO#conf t
Enter configuration commands, one per line. End with CNTL/Z.
BO(config)#int e0/0
BO(config-if)#ip nat inside
BO(config-if)#ex
BO(config)#int e0/1
BO(config-if)#ip nat outside
BO(config-if)#ex
BO(config)#access-list 10 permit 172.17.0.0 0.0.255.255
BO(config)#ip nat insode
*Sep 20 11:48:04.039: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
BO(config)#ip nt inside source list 10 interface e0/1 overload
^
% Invalid input detected at '^' marker.
BO(config)#ip nat inside source list 10 interface e0/1 overload
BO(config)#
```



TASK 2: VERIFYING INTERNET REACHABILITY USING BROWSER

TASK 3: CONFIGURING ZONES IN THE FIREWALL

The screenshot shows the Palo Alto VM (PA-VM) management interface. The left sidebar displays a network diagram with various components: DB-1, DB-2, Win1, VPC, and SW-1. The main window shows the 'Zones' configuration page. The 'Zones' table lists three zones: DMZ, inside, and outside. The 'DMZ' zone is associated with the 'ethernet1/2' interface, while 'inside' and 'outside' are associated with 'ethernet1/1' and 'ethernet1/3' respectively. The 'DMZ' zone is enabled, while 'inside' and 'outside' are disabled. The 'DMZ' zone is also associated with the 'any' user ID and 'none' device ID. The 'DMZ' zone is also associated with the 'any' user ID and 'none' device ID.

NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS
DMZ	layer3	ethernet1/2		☒		☒	any	none	☒	any	none
inside	layer3	ethernet1/1		☒		☐	any	none	☐	any	none
outside	layer3	ethernet1/3		☒		☐	any	none	☐	any	none

TASK 4: CREATING DESTINATION NAT FOR LINUX 1

The screenshot shows the Palo Alto VM (PA-VM) management interface. The left sidebar displays a network diagram with various components: DB-1, DB-2, Win1, VPC, and SW-1. The main window shows the 'Policies' configuration page. The 'Policies' table lists one rule: DNAT L1. The rule is associated with the 'outside' source zone and 'ethernet1/3' destination interface. The rule is enabled. The rule is also associated with the 'any' user ID and 'none' device ID.

NAME	TAGS	SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION
1 DNAT L1	none	outside	outside	ethernet1/3	any	PUB-IP	SSH	none

TASK 5: CREATING SECURITY POLICIES IN THE FIREWALL

The screenshot shows the Palo Alto VM (PA-VM) interface. On the left, a network diagram illustrates the setup: a 'HEAD-OFFICE' with a switch (SW-1) connected to two servers (DB-1 and DB-2) and two PCs (Win1 and VPC). The switch is connected to the PA-VM's e0/0 and e0/1 ports. The PA-VM's e0/2 port is connected to a 'BRANCH-OFFICE' switch (SW-2) which is connected to a PC (Win2). The PA-VM's e0/3 port is connected to a 'PORT-FORWARDING' switch (SW-3) which is connected to a PC (VPC). The PA-VM's e0/4 port is connected to a 'PORT-FORWARDING' switch (SW-4) which is connected to a PC (VPC).

The main window displays the 'POLICIES' tab. The 'Security' section shows a table of rules:

NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE	APPLIC
1	OUT-SERVER	none	universal	any	any	any	any	any	any	any
2	Intrazone-default	none	Intrazone	any	any	any	(Intrazone)	any	any	any
3	Interzone-default	none	Interzone	any	any	any	any	any	any	any

The 'Policy Optimizer' section shows the following status:

- No Apps Specified: 0
- Unused Apps: 0
- Rule Usage: 2
- Unused in 30 days: 2
- Unused in 90 days: 2
- Unused: 2

The bottom status bar shows the session information: admin | Logout | Last Login Time: 09/28/2026 02:05:30 | Session Expire Time: 10/28/2026 02:07:44

TASK 6: VERIFYING SSH IN OUTSIDE PC

The screenshot shows a Windows XP VM (Win2) with the following configuration:

- IP Address: 172.17.0.2
- Subnet Mask: 255.255.0.0
- Default Gateway: 172.17.0.1

The command prompt shows the following output:

```

C:\Documents and Settings\User>ipconfig

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 172.17.0.2
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 172.17.0.1

C:\Documents and Settings\User>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=2ms TTL=115
Reply from 8.8.8.8: bytes=32 time=2ms TTL=115
Reply from 8.8.8.8: bytes=32 time=2ms TTL=115
Reply from 8.8.8.8: bytes=32 time=2ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: 4, Received: 4, Sent: 4, Error: 0%
    Approximate round trip times in milliseconds:
        Minimum=0ms Maximum=0ms Average=0ms
    Internet Connection Speed:
        About 100 Mbps
  
```

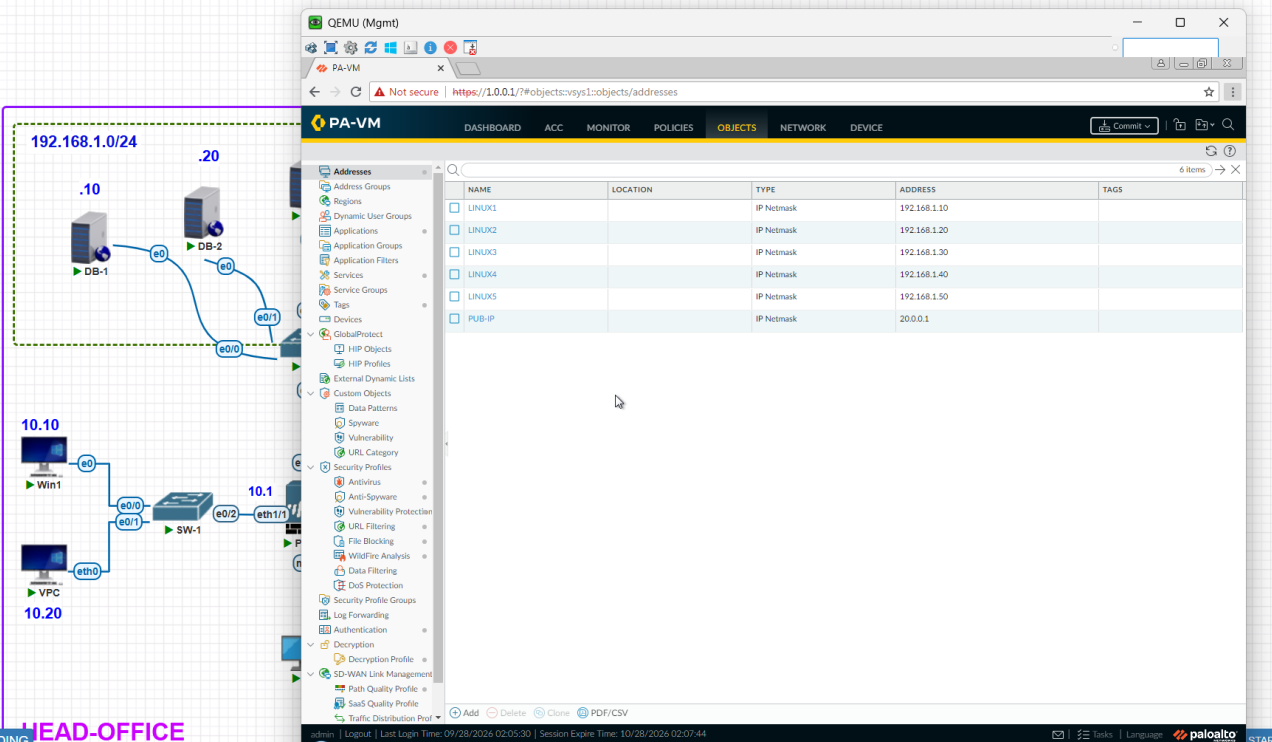
The network diagram on the left shows the 'HEAD-OFFICE' setup, including a switch (SW-1) connected to two servers (DB-1 and DB-2) and two PCs (Win1 and VPC). The switch is connected to the PA-VM's e0/0 and e0/1 ports. The PA-VM's e0/2 port is connected to a 'BRANCH-OFFICE' switch (SW-2) which is connected to a PC (Win2). The PA-VM's e0/3 port is connected to a 'PORT-FORWARDING' switch (SW-3) which is connected to a PC (VPC). The PA-VM's e0/4 port is connected to a 'PORT-FORWARDING' switch (SW-4) which is connected to a PC (VPC).

The 'BRANCH-OFFICE' section shows a switch (SW-2) connected to a PC (Win2) and a server (BO). The switch is connected to the PA-VM's e0/0 and e0/1 ports. The PA-VM's e0/2 port is connected to a 'PORT-FORWARDING' switch (SW-3) which is connected to a PC (VPC). The PA-VM's e0/3 port is connected to a 'PORT-FORWARDING' switch (SW-4) which is connected to a PC (VPC).

The bottom status bar shows the session information: Physical | 00 : 00 : 00 | START

TASK 7: ADDING ADDRESS OBJECTS IN THE FIREWALL

PORT-FORWARDING HEAD-OFFICE

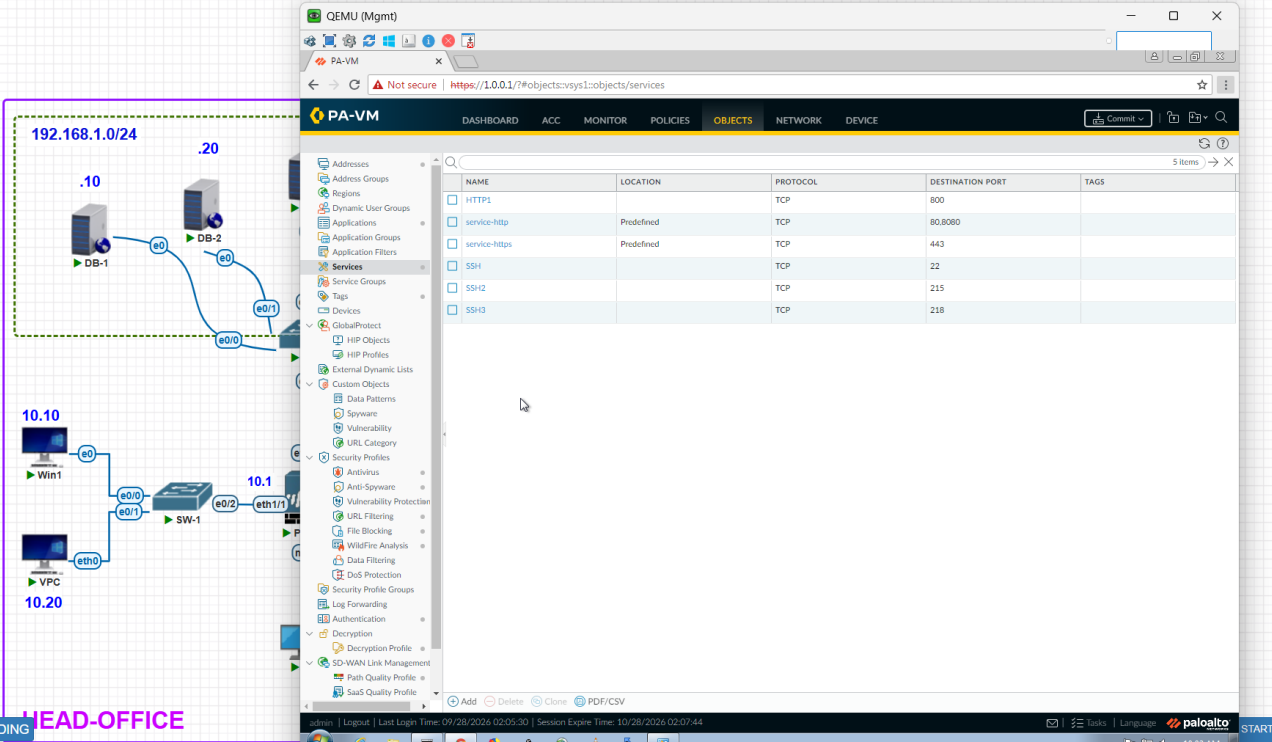


The network diagram on the left shows a topology with two main sections. The top section, enclosed in a dashed green box, contains two servers labeled DB-1 and DB-2. DB-1 has an IP of 192.168.1.0/24 and a .10 interface. DB-2 has a .20 interface. They are connected via e0 and e0/1 interfaces. The bottom section contains a switch SW-1 and two PCs, Win1 and VPC. Win1 has an IP of 10.10 and a .10 interface. VPC has an IP of 10.20 and an eth0 interface. SW-1 has interfaces e0/10, e0/11, e0/2, and eth1/1. The Palo Alto VM console on the right shows the 'OBJECTS' tab with the 'ADDRESSES' section selected. A table lists the address objects created:

NAME	LOCATION	TYPE	ADDRESS	TAGS
☐ LINUX1		IP Netmask	192.168.1.10	
☐ LINUX2		IP Netmask	192.168.1.20	
☐ LINUX3		IP Netmask	192.168.1.30	
☐ LINUX4		IP Netmask	192.168.1.40	
☐ LINUX5		IP Netmask	192.168.1.50	
☐ PUB-IP		IP Netmask	200.0.0.1	

TASK 8: CREATING SERVICE OBJECTS IN THE FIREWALL

PORT-FORWARDING HEAD-OFFICE



The network diagram on the left is identical to the one in Task 7. The Palo Alto VM console on the right shows the 'OBJECTS' tab with the 'SERVICES' section selected. A table lists the service objects created:

NAME	LOCATION	PROTOCOL	DESTINATION PORT	TAGS
☐ HTTP1		TCP	800	
☐ service-http	Predefined	TCP	80,8080	
☐ service-https	Predefined	TCP	443	
☐ SSH1		TCP	22	
☐ SSH2		TCP	215	
☐ SSH3		TCP	218	

TASK 9: CONFIGURING NAT POLICIES IN THE FIREWALL

192.168.1.0/24

.10

DB-1

e0

10.10

Win1

e0

e0/0

e0/1

SW-1

eth0

VPC

10.20

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Security

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

	NAME	TAGS	SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION
1	DNAT L1	none	outside	outside	ethernet1/3	any	PUB-IP	SSH	none
2	DNAT L2	none	outside	outside	ethernet1/3	any	PUB-IP	SSH2	none
3	DNAT L3	none	outside	outside	ethernet1/3	any	PUB-IP	SSH3	none
4	DNAT L4	none	outside	outside	ethernet1/3	any	PUB-IP	HTTP1	none
5	DNAT L5	none	outside	outside	ethernet1/3	any	PUB-IP	service-http	none

Object - Addresses

AddDeleteCloneEnableDisableMovePDF/CSVHighlight Unused RulesView Rulebase as GroupsReset Rule Hit CounterGroup

admin | Logout | Last Login Time: 09/28/2026 02:05:30 | Session Expire Time: 10/28/2026 02:07:44

TasksLanguagepal Alto00 : 00 : 00START

TASK 10: ENABLING SECURITY POLICIES

192.168.1.0/24

.10

DB-1

e0

10.10

Win1

e0

e0/0

e0/1

SW-1

eth0

VPC

10.20

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Security

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

Source			Destination			APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS
ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE					
	any	any	DMZ	any	any	any	HTTP1	Allow	none	
						any	service-http	Allow	none	
						any	SSH1	Allow	none	
						any	SSH2	Allow	none	
						any	SSH3	Allow	none	
						any	any	Deny	none	

Object - Addresses

AddDeleteCloneOverrideRevertEnableDisableMovePDF/CSVHighlight Unused RulesView Rulebase as GroupsReset Rule Hit Counter

admin | Logout | Last Login Time: 09/28/2026 02:05:30 | Session Expire Time: 10/28/2026 02:07:44

TasksLanguagepal Alto00 : 00 : 00START

TASK 11: ENABLING SSH IN THE RESPECTIVE SERVERS

192.168.1.0/24

DB-1

```

4 bytes from 192.168.1.50: seq=0 ttl=64 time=2.859 ms
4 bytes from 192.168.1.50: seq=1 ttl=64 time=1.037 ms
4 bytes from 192.168.1.50: seq=2 ttl=64 time=1.001 ms
4 bytes from 192.168.1.50: seq=3 ttl=64 time=1.105 ms
--- 192.168.1.50 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.001/1.500/2.859 ms

ssh root@192.168.1.10

Host '192.168.1.10' is not in the trusted hosts file.
(fingerprint md5 86:29:e4:bd:32:1c:21:7d:a0:1f:2e:65:23:f2:6b:81)
Do you want to continue connecting? (y/n) y
root@192.168.1.10's password:
Chop wood, carry water.
  
```

DB-2

```

round-trip min/avg/max = 1.804/2.262/3.176 ms
ping 192.168.1.50
PING 192.168.1.50 (192.168.1.50): 56 data bytes
64 bytes from 192.168.1.50: seq=0 ttl=64 time=2.025 ms
64 bytes from 192.168.1.50: seq=1 ttl=64 time=1.108 ms
64 bytes from 192.168.1.50: seq=2 ttl=64 time=1.625 ms
64 bytes from 192.168.1.50: seq=3 ttl=64 time=1.489 ms
--- 192.168.1.50 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.188/1.781/2.825 ms

ssh root@192.168.1.20

Host '192.168.1.20' is not in the trusted hosts file.
(fingerprint md5 86:29:e4:bd:32:1c:21:7d:a0:1f:2e:65:23:f2:6b:81)
Do you want to continue connecting? (y/n) y
root@192.168.1.20's password:
Chop wood, carry water.
  
```

DB-3

```

round-trip min/avg/max = 1.591/2.622/5.006 ms
ping 192.168.1.40
PING 192.168.1.40 (192.168.1.40): 56 data bytes
64 bytes from 192.168.1.40: seq=0 ttl=64 time=4.990 ms
64 bytes from 192.168.1.40: seq=1 ttl=64 time=1.349 ms
64 bytes from 192.168.1.40: seq=2 ttl=64 time=0.995 ms
64 bytes from 192.168.1.40: seq=3 ttl=64 time=2.137 ms
--- 192.168.1.40 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.995/2.367/4.990 ms

ssh root@192.168.1.10

Host '192.168.1.10' is not in the trusted hosts file.
(fingerprint md5 86:29:e4:bd:32:1c:21:7d:a0:1f:2e:65:23:f2:6b:81)
Do you want to continue connecting? (y/n) y
root@192.168.1.10's password:
Chop wood, carry water.
  
```

1.0.0.10

PORT-FORWARDING HEAD-OFFICE

SUREN NETWORK
PLAY WITH TECHNOLOGIES

TASK 12: VERIFYING SSH FROM THE BRANCH OFFICE PC WITH CUSTOMIZED PORT NUMBER

192.168.1.0/24

DB-1

```

Microsoft Windows [Version 6.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 
    Subnet Mask . . . . . : 
    Default Gateway . . . . . : 

C:\Documents and Settings\User>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=2ms TTL=64
Reply from 8.8.8.8: bytes=32 time=2ms TTL=64
Reply from 8.8.8.8: bytes=32 time=3ms TTL=64

Ping statistics for 8.8.8.8:
    Packets: 3, Received: 3, Sent: 3, Error: 0%
    Round-trip times:
        Minimum: 2 ms, Maximum: 3 ms, Average: 2 ms
  
```

20.0.0.1 - PuTTY

```

User as: root
root@20.0.0.1's password:
Chop wood, carry water.
  
```

20.0.0.1 - PuTTY

```

User as: root
root@20.0.0.1's password:
Chop wood, carry water.
  
```

20.0.0.1 - PuTTY

```

User as: root
root@20.0.0.1's password:
Chop wood, carry water.
  
```

10.10

10.20

BRANCH-OFFICE

172.17.0.10/16

30.2

172.17.0.1

PORT-FORWARDING HEAD-OFFICE

SUREN NETWORK
PLAY WITH TECHNOLOGIES

TASK 13: VERIFYING HTTP FROM THE BRANCH OFFICE PC WITH CUSTOMIZED PORT NUMBER

